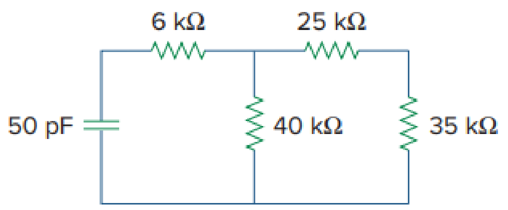
Chapter 7, Problem 3P

Problem

Determine the time constant for the circuit in Fig. 7.83.

Fig. 7.83:



Step-by-step solution

Step 1 of 4

The expression for the time constant of the circuit is,

$\tau = R_{eq}C$

Here R_{eq} is the equivalent resistance seen by the capacitor.

Consider the circuit in Figure 1 to calculate R_{eq}.

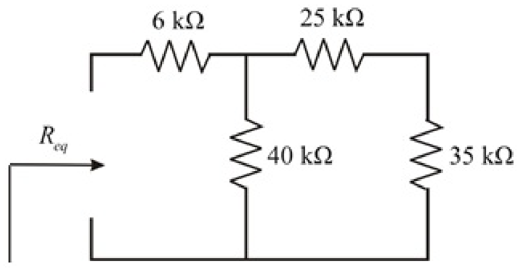


Figure 1

Step 2 of 4

From Figure 1, $25\text{ k}\Omega$ and $35\text{ k}\Omega$ resistors are in series. Hence the equivalent resistance is,

$$\begin{aligned} R_{eq1} &= 25\text{ k}\Omega + 35\text{ k}\Omega \\ &= 60\text{ k}\Omega \end{aligned}$$

The circuit is reduced to as shown in Figure 2.

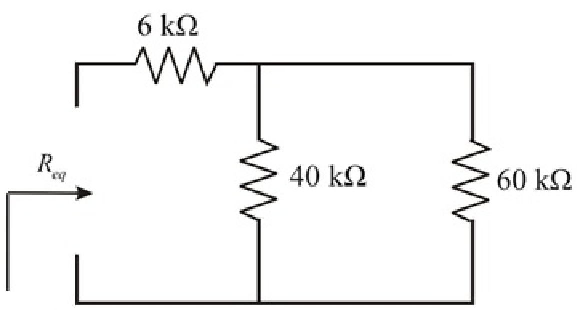


Figure 2

Step 3 of 4

From Figure 2, $60\ \text{k}\Omega$ and $40\ \text{k}\Omega$ are in parallel.

Hence,

$$R_{eq2} = \frac{60\text{k} \times 40\text{k}}{60\text{k} + 40\text{k}}$$
$$= 24\ \text{k}\Omega$$

The circuit is reduced to as shown in Figure 3.

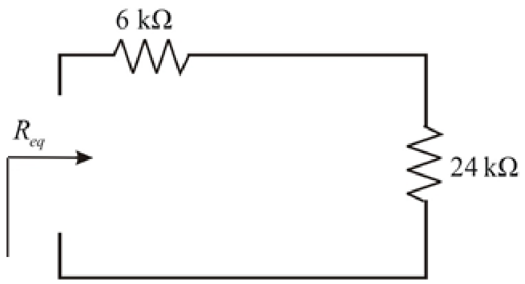


Figure 3

Step 4 of 4

From Figure 3, the $6\ \text{k}\Omega$ and $24\ \text{k}\Omega$ resistors are in parallel. Hence the equivalent resistnce seen by the capacitor is,

$$R_{eq} = 24\ \text{k}\Omega + 6\ \text{k}\Omega$$
$$= 30\ \text{k}\Omega$$

The time constant of the circuit is,

$$\tau = R_{eq}C$$

Substitute $30\ \text{k}\Omega$ for R_{eq} and 50 pF for *C*.

$$\tau = \left(50\times10^{-12}\right)\left(30\times10^{3}\right)$$
$$= 1.5\times10^{-6}$$
$$= 1.5\ \mu\text{sec}$$

Thus, the time constant of the circuit is $\boxed{1.5\ \mu\text{sec}}$.